

Make notes Very detailed (full sentences, no short phrases)

- ✓ Including all important theoretical points
- ✓ Including procedural steps wherever steps are given
- ✓ Formatted cleanly for exam

make sure to include each and every imp point.



Below are **very detailed, exam-ready notes** from **AWS Academy Cloud Foundations – Module 6: Compute**, written fully in sentences (no short phrases), clearly structured, and covering **all important theoretical concepts and procedures** from the module.

(Reference source: AWS Academy Module 6 PDF  AcademyCloudFoundations_Module_...)

MODULE 6 – COMPUTE (AWS CLOUD FOUNDATIONS)

✓ 1. OVERVIEW OF AWS COMPUTE SERVICES

Amazon Web Services provides multiple compute services that allow users to run applications in different ways depending on control, scalability, and management requirements.

The major compute service categories include:

➤ Infrastructure as a Service (IaaS) – Amazon EC2

Amazon EC2 provides virtual machines called instances that users fully control. Users manage the operating system, software, and configuration.

➤ Serverless Computing – AWS Lambda

AWS Lambda runs code without managing servers.

It executes only when triggered by events and charges only for execution time.

➤ Container-Based Computing – Amazon ECS, Amazon EKS, AWS Fargate

These services allow applications to run inside containers rather than full virtual machines. AWS Fargate reduces server management overhead.

➤ Platform as a Service (PaaS) – AWS Elastic Beanstalk

Elastic Beanstalk automatically manages infrastructure while users focus only on application code.

✅ 2. AMAZON EC2 (ELASTIC COMPUTE CLOUD)

Amazon EC2 provides scalable virtual servers in the cloud known as EC2 instances.

◆ Key Features of Amazon EC2:

Amazon EC2 allows users to:

- Run Windows and Linux virtual machines
- Launch instances from preconfigured templates called AMIs
- Choose instance size based on CPU, memory, storage, and network needs
- Control inbound and outbound traffic
- Deploy instances globally within Availability Zones

EC2 instances can act as:

Application servers, web servers, database servers, game servers, mail servers, media servers, file servers, and computing servers.

✅ 3. STEPS TO LAUNCH AN AMAZON EC2 INSTANCE (9 DECISIONS)

🧩 Step 1: Select an Amazon Machine Image (AMI)

An AMI is a template used to create an EC2 instance.

It includes:

- Operating system (Linux or Windows)
- Preinstalled software (optional)

Types of AMIs:

- Quick Start AMIs provided by AWS
- My AMIs created by users
- AWS Marketplace AMIs from vendors
- Community AMIs shared by others

You can also create a custom AMI by:

Launching an instance → modifying it → capturing it as a new AMI → optionally copying it to other regions.

Step 2: Select an Instance Type

The instance type determines:

- CPU power
- Memory (RAM)
- Storage capacity and type
- Network performance

Instance categories include:

Category	Use
General Purpose	Balanced workloads
Compute Optimized	High CPU workloads
Memory Optimized	Large in-memory apps
Storage Optimized	High disk usage
Accelerated Computing	Machine learning & graphics

Example:

`t3.large` means:

- T = family
 - 3 = generation
 - large = size
-

Step 3: Specify Network Settings

You must choose:

- A Virtual Private Cloud (VPC)
- A subnet within an Availability Zone
- Whether to assign a public IP address

Public IP allows internet access.

Step 4: Attach an IAM Role (Optional)

An IAM role gives EC2 permission to interact with other AWS services.

Example:

An EC2 instance can access Amazon S3 without storing credentials.

Roles can be attached at launch or later.

Step 5: Add User Data Script (Optional)

User data allows you to run a script when the instance starts for the first time.

Uses include:

- Installing software
- Updating OS packages
- Automating setup tasks

This reduces the need for many custom AMIs.

Step 6: Specify Storage

You configure:

- Root volume (OS disk)
- Additional volumes (optional)

For each volume you select:

- Size

- Storage type (SSD or HDD)
 - Deletion on termination
 - Encryption
-

Step 7: Add Tags

Tags are key-value pairs used to label resources.

Benefits include:

- Easy filtering
- Cost tracking
- Automation
- Access control

Example:

Key: Name

Value: WebServer1

Step 8: Configure Security Group

A security group acts as a firewall.

It defines:

- Allowed ports
- Protocols (TCP, UDP, ICMP)
- Source IPs or security groups

Example:

Allow HTTP traffic on port 80 from anywhere.

Step 9: Create or Select Key Pair

A key pair enables secure login.

It includes:

- Public key stored by AWS
- Private key stored by user

For Linux → SSH login

For Windows → decrypt administrator password

✓ 4. EC2 STORAGE OPTIONS

Amazon Elastic Block Store (EBS)

- Persistent block storage
- Data remains even when instance stops
- Used for root volumes

EC2 Instance Store

- Temporary storage
- Data lost when instance stops or fails

Other Options

- Amazon EFS for file systems
 - Amazon S3 for object storage
-

✓ 5. EC2 INSTANCE LIFECYCLE

Main states include:

- Pending → Running
- Stopping → Stopped
- Shutting down → Terminated

Only **EBS-backed instances** can be stopped.

✓ 6. ELASTIC IP ADDRESS

Stopping and starting an instance changes its public IP.

Elastic IP provides:

- Permanent public IP
- Can be attached and detached
- Remains until released

Used when a fixed IP is required.

7. EC2 METADATA

Metadata provides information about a running instance such as:

- IP addresses
- Instance ID
- Availability Zone
- Security groups

Accessible inside the instance using:

<http://169.254.169.254/latest/meta-data/> ↗

Used for automation and configuration.

8. EC2 MONITORING WITH CLOUDWATCH

Amazon CloudWatch provides:

Basic Monitoring

- Free
- Metrics every 5 minutes

Detailed Monitoring

- Paid
- Metrics every 1 minute

It stores up to 15 months of data.

EC2 COST OPTIMIZATION

◆ EC2 PRICING MODELS

✓ On-Demand Instances

Pay per hour or second with no commitment.

✓ Reserved Instances

Commit for 1 or 3 years for discounted pricing.

✓ Spot Instances

Use spare capacity at low cost but can be interrupted.

✓ Dedicated Hosts

Physical servers dedicated to one customer.



FOUR PILLARS OF COST OPTIMIZATION

1. Right Sizing

Provision instances that match CPU, memory, storage, and network needs using CloudWatch metrics.

2. Increase Elasticity

Stop unused instances and use Auto Scaling to match demand.

3. Optimal Pricing Model

Combine On-Demand, Spot, and Reserved Instances based on workload.

4. Optimize Storage

Resize volumes, change storage types, delete unused snapshots, and use Amazon S3 lifecycle policies.



AWS LAMBDA (SERVERLESS COMPUTING)

AWS Lambda runs code without servers.

◆ Key Characteristics:

- Code runs only when triggered

- Automatically scales
- Charges only for execution time

Event Sources Include:

- Amazon S3
 - DynamoDB
 - SNS
 - SQS
 - API Gateway
 - CloudWatch
-

Benefits of Lambda:

- Supports many programming languages
 - Fully managed infrastructure
 - Built-in fault tolerance
 - Pay-per-use pricing
 - Supports function orchestration
-

Lambda Limits:

- Maximum memory: 10,240 MB
 - Maximum runtime: 15 minutes
 - Concurrent executions: 1,000 per region
-

AWS ELASTIC BEANSTALK

Elastic Beanstalk is a managed service for deploying web applications.

It automatically handles:

- Infrastructure provisioning
- Load balancing
- Auto scaling
- Monitoring

- Logging
- Health checks

Supported languages:

Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker.

Pricing:

There is **no extra charge** for Elastic Beanstalk.

You pay only for AWS resources used.

BENEFITS OF ELASTIC BEANSTALK

- Fast deployment
 - High developer productivity
 - Reduced management complexity
 - Full resource control
-

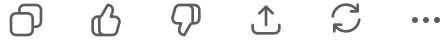
FINAL MODULE TAKEAWAYS

- ✓ AWS provides multiple compute models: EC2, Lambda, containers, and Elastic Beanstalk
 - ✓ EC2 offers full control over virtual machines
 - ✓ EC2 launch involves AMI, instance type, networking, storage, security, and access
 - ✓ Cost optimization relies on right sizing, elasticity, pricing models, and storage efficiency
 - ✓ Lambda enables serverless event-driven computing
 - ✓ Elastic Beanstalk simplifies web application deployment
-

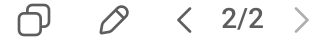
If you'd like, I can next:

- ✓ Convert these into **short revision notes**
- ✓ Create **exam questions + answers**
- ✓ Make **memory tricks for EC2 & pricing models**
- ✓ Turn into **one-page cheat sheet**

Just tell me 



memory tricks for learning all the most imp stuff only from this module



Perfect 👍 Let's make this **super exam-focused memory tricks sheet** for Module 6 (Compute).

Only the **most important, repeatedly asked concepts** are included.

🧠 MODULE 6 – MEMORY TRICKS (EXAM FOCUSED)

1 Types of Compute Services

🔑 Trick: "I Serve Containers Properly"

- **I** → IaaS → Amazon EC2
- **Serve** → Serverless → AWS Lambda
- **Containers** → ECS, EKS, Fargate
- **Properly** → PaaS → Elastic Beanstalk

👉 If question asks: *Which service gives full control?* → EC2

👉 If question asks: *No server management?* → Lambda

👉 If question asks: *Deploy web app quickly?* → Elastic Beanstalk

2 9 Steps to Launch EC2 (VERY IMPORTANT)

🔑 Trick: "Aunt Is Never In Uniform, So She Sings Kindly"

Each first letter = one EC2 launch step:

Word	Step
Aunt	AMI
Is	Instance Type
Never	Network
In	IAM Role
Uniform	User Data
So	Storage
She	Security Group
Sings	Tags
Kindly	Key Pair

If you remember the sentence, you remember all 9 steps in order.

Instance Type Categories

Trick: “General Computers Make Smart Apps”

- **General** → General purpose
- **Computers** → Compute optimized
- **Make** → Memory optimized
- **Smart** → Storage optimized
- **Apps** → Accelerated computing

If exam gives workload:

- High CPU → Compute optimized
 - Big database → Memory optimized
 - Large disk workload → Storage optimized
 - ML / GPU → Accelerated
-

4 Instance Type Naming (t3.large)

🔑 Trick: “Family → Generation → Size”

Example: t3.large

- t = family
- 3 = generation
- large = size

Think: “Last name, birth year, body size.”

5 Storage Types (Very Important)

🔑 Trick: “EBS Exists, Instance Store Vanishes”

- **EBS** → Data stays after stop
- **Instance Store** → Data disappears if stopped

If question says:

Instance stops due to failure, what happens to data?

If EBS → Safe

If Instance Store → Lost

6 Only EBS-backed Instances Can Be Stopped

🔑 Trick:

“If you want to STOP, choose EBS on TOP.”

Instance Store instances cannot be stopped.

7 Security Group

🔑 Trick:

“Security Group = Firewall Outside OS”

Remember:

- Controls ports
- Controls protocol
- Controls source

Always think:

Port + Protocol + Source

Elastic IP

 **Trick:**

“Stop = IP Swap”

If you stop and start an instance:

- Public IP changes
- Private IP stays same

If fixed public IP needed → Use Elastic IP

EC2 Pricing Models (Very Important)

 **Trick: “On Rare Special Days”**

- **On** → On-Demand
 - **Rare** → Reserved
 - **Special** → Spot
 - **Days** → Dedicated
-

When to Use Each?

 **Trick: “Spiky? Spot it. Stable? Reserve it.”**

- Short-term → On-Demand
- Predictable workload → Reserved

- Flexible workload → Spot
 - Compliance/BYOL → Dedicated Host
-

10 Four Pillars of Cost Optimization

Trick: “RICE”

- **R** → Right size
- **I** → Increase elasticity
- **C** → Correct pricing model
- **E** → Efficient storage

If question asks cost optimization → Think **RICE**

1 1 AWS Lambda Basics

Trick:

“Lambda = No Server, Only Code.”

Remember 3 main things:

- Runs when triggered
 - Auto scales
 - Pay per execution
-

Lambda Limits (Very Exam Important)

Trick: “10 – 15 – 1000 Rule”

- 10 GB max memory (10,240 MB)
- 15 minutes max runtime
- 1000 concurrent executions

If you remember 10–15–1000 → You get Lambda limit questions right.

1 2 Elastic Beanstalk

🔑 Trick:

“Beanstalk Builds While You Code.”

It automatically handles:

- Infrastructure
- Scaling
- Load balancing
- Monitoring

You only upload code.

If exam asks:

Which service helps developers quickly deploy apps in Java, .NET etc?

Answer → Elastic Beanstalk

🔥 EC2 vs RDS Trick (Common Question)

🔑 Trick:

“RDS Relaxes, EC2 Effort.”

- RDS = Managed (AWS handles patching, backups)
 - EC2 = You manage everything
-

⚡ Quick Comparison Memory Table

If Question Says...

Answer

Full OS control

EC2

No server management

Lambda

If Question Says...	Answer
Web app easy deploy	Elastic Beanstalk
Cheap but interruptible	Spot
Fixed long-term workload	Reserved
Temporary storage	Instance Store
Persistent storage	EBS
Firewall control	Security Group

Ultra-Short Revision Formula

If you remember just this, you can pass:

- 9 EC2 steps → “Aunt Is Never In Uniform So She Sings Kindly”
- Pricing → “On Rare Special Days”
- Cost Pillars → “RICE”
- Lambda limits → “10–15–1000”
- Storage → “EBS Exists, Instance Store Vanishes”
- Elastic Beanstalk → “You code, AWS builds”

If you want next, I can give you:

- ✓ 15 Most Expected Exam Questions
- ✓ Super Short 2-Page Final Revision Sheet
- ✓ Tricky MCQs from this module
- ✓ Viva preparation answers

Tell me what you need 

